

Exercises 1-2 Answer

5.

pf.

Sufficiency: Given that $\{a_{2k}\}$ and $\{a_{2k-1}\}$ converges to a .

$$\begin{aligned}\forall \varepsilon > 0, \exists K_1 \text{ s.t. } k > K_1, |a_{2k} - a| < \varepsilon, \\ \exists K_2 \text{ s.t. } k > K_2, |a_{2k-1} - a| < \varepsilon.\end{aligned}$$

Let $N = \max\{K_1, K_2\}$, $n > N$, $\forall \varepsilon$:

$$\begin{aligned}n = 2k, |a_n - a| &= |a_{2k} - a| < \varepsilon, \\ n = 2k - 1, |a_n - a| &= |a_{2k-1} - a| < \varepsilon. \\ \Rightarrow |a_n - a| < \varepsilon &\Rightarrow \lim_{n \rightarrow \infty} a_n = a.\end{aligned}$$

Necessity: Given that $\{a_n\}$ converges to a .

By definition:

$$\begin{aligned}\forall \varepsilon, \exists N, \text{ s.t. } n > N, |a_n - a| < \varepsilon : \\ n = 2k, |a_{2k} - a| &= |a_n - a| < \varepsilon, \\ n = 2k - 1, |a_{2k-1} - a| &= |a_n - a| < \varepsilon. \\ \Rightarrow \lim_{n \rightarrow \infty} a_{2k} = a, \lim_{n \rightarrow \infty} a_{2k-1} &= a. \blacksquare\end{aligned}$$

6.

$$\lim_{n \rightarrow \infty} \left| \frac{1}{n} - \frac{2}{n} + \frac{3}{n} + \dots + \frac{(-1)^{n-1}n}{n} \right|.$$

$$\text{let } a_n = \frac{1}{n} - \frac{2}{n} + \frac{3}{n} + \dots + \frac{(-1)^{n-1}n}{n}.$$

$$\begin{aligned}a_{2k} &= \frac{1}{2k} - \frac{2}{2k} + \frac{3}{2k} + \dots + \frac{2k-1}{2k} - \frac{2k}{2k} \\ &= \left(-\frac{1}{2k} \right) \cdot k \\ &= -\frac{1}{2},\end{aligned}$$

$$\lim_{n \rightarrow \infty} |a_{2k}| = \frac{1}{2}.$$

$$\begin{aligned}a_{2k-1} &= \frac{1}{2k-1} - \frac{2}{2k-1} + \frac{3}{2k-1} + \dots + \frac{2k-3}{2k-1} - \frac{2k-2}{2k-1} + \frac{2k-1}{2k-1} \\ &= \left(-\frac{1}{2k-1} \right) \cdot (k-1) + 1\end{aligned}$$

$$\lim_{n \rightarrow \infty} |a_{2k-1}| = \lim_{n \rightarrow \infty} \left| \frac{k}{2k-1} \right| = \frac{1}{2}.$$

$$\therefore [5],$$

$$\therefore \lim_{n \rightarrow \infty} |a_n| = \frac{1}{2}.$$

8.

(2)

$$\lim_{n \rightarrow \infty} \sqrt[n]{a^n + b^n}.$$

WLOG, assume that $0 < a \leq b$.

$$\begin{aligned} \sqrt[n]{b^n} &< \sqrt[n]{a^n + b^n} \leq \sqrt[n]{2b^n}, \\ \lim_{n \rightarrow \infty} (\sqrt[n]{2b^n}) &= \lim_{n \rightarrow \infty} \sqrt[n]{2} \times \lim_{n \rightarrow \infty} \sqrt[n]{b^n} = 1 \times b = b, \\ \lim_{n \rightarrow \infty} \sqrt[n]{b^n} &= b. \\ \Rightarrow \lim_{n \rightarrow \infty} \sqrt[n]{a^n + b^n} &= \max\{a, b\}. \end{aligned}$$

(3)

$$\lim_{n \rightarrow \infty} \sqrt[n]{n^p + n^q}, \quad p, q \in \mathbb{N}^*.$$

WLOG, assume that $0 < p \leq q$.

$$\begin{aligned} \sqrt[n]{n^q} &< \sqrt[n]{n^p + n^q} \leq \sqrt[n]{2n^q}, \\ \lim_{n \rightarrow \infty} (\sqrt[n]{n^q}) &= \left(\lim_{n \rightarrow \infty} \sqrt[n]{n} \right)^q = 1^q = 1, \\ \lim_{n \rightarrow \infty} \sqrt[n]{2n^q} &= \lim_{n \rightarrow \infty} \sqrt[n]{2} \times \lim_{n \rightarrow \infty} \sqrt[n]{n^q} = 1 \times 1 = 1. \\ \Rightarrow \lim_{n \rightarrow \infty} \sqrt[n]{n^p + n^q} &= 1. \end{aligned}$$

(4)

$$\begin{aligned} \lim_{n \rightarrow \infty} [(n+1)^\alpha - n^\alpha], \quad 0 < \alpha < 1. \\ \lim_{n \rightarrow \infty} [(n+1)^\alpha - n^\alpha] \\ &= \lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n}\right)^\alpha - 1 \right] n^\alpha. \\ \because 0 < \left(1 + \frac{1}{n}\right)^\alpha &\leq 1 + \frac{\alpha}{n} - 1 = \frac{\alpha}{n} \quad [\text{Bernoulli's inequality}], \\ \therefore 0 < (n+1)^\alpha - n^\alpha &\leq \alpha n^{\alpha-1}. \\ \lim_{n \rightarrow \infty} \alpha n^{\alpha-1} &= 0. \\ \Rightarrow \lim_{n \rightarrow \infty} [(n+1)^\alpha - n^\alpha] &= 0. \end{aligned}$$

(5)

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt[n]{n!}}.$$

$$(1.1)^n < n! < n^n, n \geq 2.$$

$$\frac{1}{\sqrt{n^n}} < \frac{1}{\sqrt{n!}} < \frac{1}{(1.1)^{\frac{n}{2}}},$$

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n^n}} = 0,$$

$$\lim_{n \rightarrow \infty} \frac{1}{(1.1)^{\frac{n}{2}}} = 0.$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n!}} = 0.$$

9.

(1)

$$x_n = \frac{a^n}{n!} (a > 0).$$

$$\frac{x_{n+1}}{x_n} = \frac{a}{n+1},$$

Let $N = [a - 1] + 1$,

$$\forall n > N, \frac{x_{n+1}}{x_n} < 1.$$

$\{x_n\}$ decreases, while $x_n = \frac{a^n}{n!} > 0$, $\{x_n\}$ converges to 0.

$$\lim_{n \rightarrow \infty} x_n = 0$$

(2)

$$a > 0, x_0 > 0, k \in \mathbb{N}^*, x_n > 0,$$

$$x_{n+1} = \frac{1}{k+1} \left(kx_n + \frac{a}{x_n^k} \right)$$

Use Am-Gm inequality:

$$x_{n+1} = \frac{1}{k+1} \left(\underbrace{x_n + x_n + \cdots + x_n}_k + \frac{1}{x_n^k} \right) \geq {}^{k+1}\sqrt{a}.$$

$$\begin{aligned} x_{n+1} - x_n &= -\frac{1}{k+1}x_n + \frac{a}{(k+1)x_n^k} \\ &= \frac{1}{k+1} \left(\frac{a}{x_n^k} - x_n \right) \\ &= \frac{x_n}{k+1} \left(\frac{a}{x_n^{k+1}} - 1 \right) \\ &\leq \frac{x_n}{k+1} \left(\frac{a}{({}^{k+1}\sqrt{a})^{k+1}} - 1 \right) = 0. \end{aligned}$$

Thus, $\{x_n\}$ decreases. Let $\lim_{n \rightarrow \infty} x_n = L$.

$$L = \frac{1}{k+1} \left(kL + \frac{a}{L^k} \right)$$
$$\Rightarrow \lim_{n \rightarrow \infty} x_n = L = \sqrt[k+1]{a}.$$

10.

pf.

$\{x_n\}$ increases and $\{y_n\}$ decreases, so $\{x_n - y_n\}$ increases.

$$\begin{aligned} \because \lim_{n \rightarrow \infty} (x_n - y_n) &= 0, \\ \therefore x_1 - y_1 &\leq x_n - y_n < 0 \text{ [sign preservation]} \\ &\Rightarrow x_n < y_n. \end{aligned}$$

If $\{x_n\}$ and $\{y_n\}$ are unbounded,

$$\begin{aligned} \forall M_1, \exists N_1, \forall n > N_1, x_n > M_1, \\ \forall M_2, \exists N_2, \forall n > N_2, y_n < M_2. \end{aligned}$$

Thus, for $n > \max\{N_1, N_2\}$,

$$x_n - y_n > M_1 - M_2 = M, \forall M.$$

Then $\{x_n - y_n\}$ diverges, a contradiction. Thus, $\{x_n\}$ or $\{y_n\}$ is bounded.

If $\{x_n\}$ is bounded, by MCT, $\lim_{n \rightarrow \infty} x_n$ exists, and

$$x_1 \leq x_n \leq \lim_{n \rightarrow \infty} x_n.$$

If $\{y_n\}$ is unbounded,

$$x_n - y_n > x_1 - M_2, \forall M_2,$$

which gives a contradiction. Thus, $\lim_{n \rightarrow \infty} y_n$ exists.

Therefore, both $\lim_{n \rightarrow \infty} x_n$ and $\lim_{n \rightarrow \infty} y_n$ exist.

$$\begin{aligned} \lim_{n \rightarrow \infty} (x_n - y_n) &= \lim_{n \rightarrow \infty} x_n - \lim_{n \rightarrow \infty} y_n = 0. \\ &\Rightarrow \lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} y_n. \blacksquare \end{aligned}$$

12.

pf.

If $\{x_n\}$ converges to $L > 0$,

$$\begin{aligned} L = \sqrt{b + L} &\Rightarrow L^2 - L - b = 0, \\ &\Rightarrow L = \frac{1 + \sqrt{1 + 4b}}{2}. \end{aligned}$$

(i) If $x_1 = a < L$, assume that $x_n < L$.

$$\begin{aligned} x_{n+1} &= \sqrt{b + x_n} < \sqrt{b + L} = \sqrt{L^2} = L, \\ x_{n+1} - x_n &= \sqrt{b + x_n} - x_n > \sqrt{b + x_n} - L > 0. \end{aligned}$$

So $\{x_n\}$ increases and is bounded.

$$\Rightarrow \lim_{n \rightarrow \infty} x_n = L.$$

(ii) If $x_1 = a > L$, assume that $x_n > L$.

$$x_{n+1} = \sqrt{b + x_n} > \sqrt{b + L} = \sqrt{L^2} = L,$$

$$x_n^2 - x_n - L > 0 \Rightarrow x_n > \sqrt{b + x_n} = x_{n+1}.$$

So $\{x_n\}$ decreases and is bounded.

$$\Rightarrow \lim_{n \rightarrow \infty} x_n = L.$$

(iii) If $x_1 = L$, assume that $x_n = L$.

$$x_{n+1} = \sqrt{b + L} = L.$$

Obviously,

$$\lim_{n \rightarrow \infty} x_n = L. \blacksquare$$

15.

pf.

Because $\{x_{n_k}\}$ is bounded,

$$\exists M, \forall k, |x_{n_k}| \leq M.$$

$$\forall n, \exists k, x_n < x_{n_k} \leq M.$$

Thus, $\{x_n\}$ is bounded. Since $\{x_n\}$ is increasing, $\{x_n\}$ converges. $\blacksquare$

Exercises 1-3 Answer

1.

(3) **pf.**

Because $x \neq -1$,

$$\lim_{x \rightarrow -1} \frac{x+1}{(x+1)(x-1)} = \lim_{x \rightarrow -1} \frac{1}{x-1}.$$

For $\forall \varepsilon > 0$, let $\delta = \min\{1, 2\varepsilon\}$. When $0 < |x+1| < \delta$,

$$\left| \frac{1}{x-1} + \frac{1}{2} \right| = \frac{|x+1|}{2|x-1|} < \frac{|x+1|}{2} < \varepsilon. \blacksquare$$

(5) **pf.**

$$\left| \frac{1}{x^2} \sin x \right| \leq \left| \frac{1}{x^2} \right| < \varepsilon \Rightarrow x > \sqrt{\frac{1}{\varepsilon}}. \blacksquare$$

Let $X = \frac{1}{\sqrt{\varepsilon}}$.

$$\forall \varepsilon > 0, x > X, \left| \frac{1}{x^2} \sin x \right| < \varepsilon.$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{1}{x^2} \sin x = 0. \blacksquare$$

4.

$$f(x) = \begin{cases} x+a, & x < 0 \\ x^3+2, & x \geq 0 \end{cases}$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (x+a) = a,$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (x^3+2) = 2.$$

Because $\lim_{x \rightarrow 0} f(x)$ exists,

$$a = 2.$$